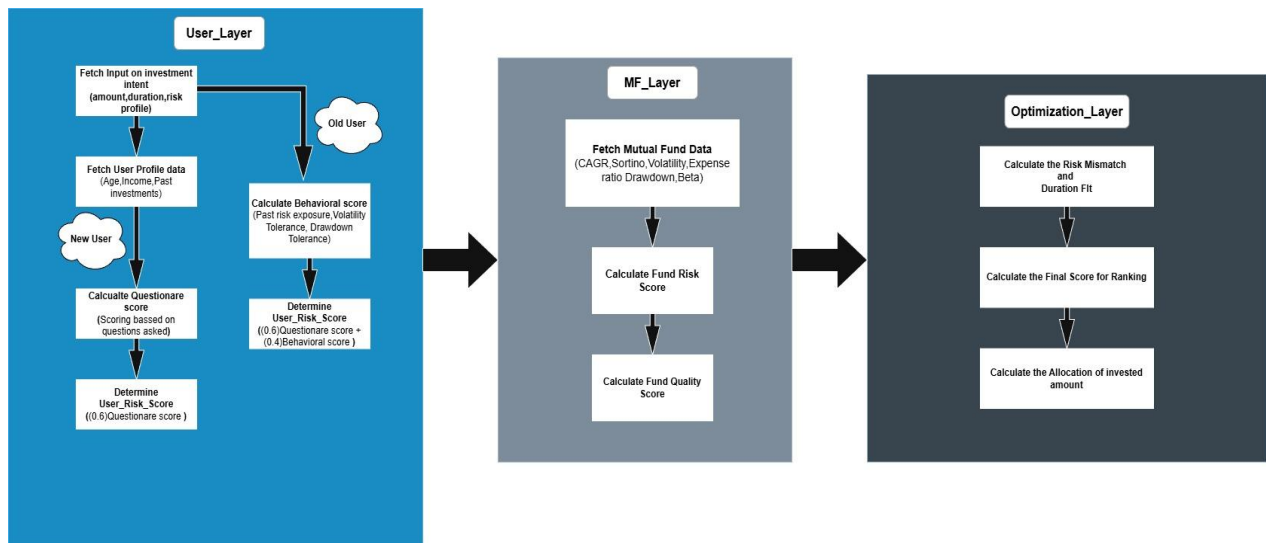


Project: Purchase Optimizer for Mutual Funds



Layer1(USER):

Investment Input:

1. Amount
2. Duration
3. Risk profile

User details

1. Age
2. Income
3. Past Investments

Formulas:

1.Normalisation formula

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min})$$

Note: Use min-max normalization across all funds.

2.User Risk Score formula

Risk Capacity Score:

1. $Age_score = 1 - (Age / 60)$
2. $Income_score = \min(income / 2000000, 1)$
3. $Duration_score = \min(duration / 15, 1)$

All values will have to be normalised

$$\text{Risk_Capacity_Score} = (0.4) * \text{Age_score} + (0.3) * \text{Income_score} + (0.3) * \text{Duration_score}$$

Stated Risk Score:

The Questionnaire score will be calculated based on the 4 psychological multiple choice questions where the input of the user will help in determining the Stated_RiskScore.

$$\text{Questionnaire_Score} = (Q1 + Q2 + Q3 + Q4) / 4$$

$$\text{Stated_RiskScore} = (\text{Questionnaire_Score} - 1) / 3$$

For new User

$$\text{User_risk_score} = 0.6 * \text{Stated_RiskScore} + 0.4 * \text{Risk_Capacity}$$

For User with past investments

$$1. \text{Past_Risk_Exposure} = \sum w_i * \text{FundRiskScore}_i$$

$$2. \text{Vol_Tolerance} = \text{UserPortfolio_Vol} / \text{Market_Vol}$$

$$3. \text{Drawdown_Tolerance} = \text{Endured_Drawdown} / \text{Market_Max_Drawdown}$$

$$\text{RevealedRiskScore} = (0.5) * \text{Past_Risk_Exposure} + (0.3) * \text{Vol_Tolerance} + (0.2) * \text{Drawdown_Tolerance}$$

$$1. \alpha = \max(0.2, e^{-\lambda * \text{history_years}})$$

$$2. \text{History_years} = \sum (\text{Months invested} * \text{Capital_weight}) / 12$$

$$\text{User_Risk_Score} = \alpha * \text{Stated_RiskScore} + (1 - \alpha) * \text{RevealedRiskScore}$$

Layer 2(MF)

1. Log Returns:

$$R_t = \ln(\text{NAV}_t / \text{NAV}_{t-1})$$

$$R^b_t = \ln(\text{Index}_t / \text{Index}_{t-1})$$

2. Volatility:

$$\sigma = \sqrt{(\sum (R_t - \bar{R})^2 / (N - 1))}$$

$$\text{Volatility} = \sigma \times \sqrt{252}$$

3. Maximum Drawdown:

$$\text{Peak}_t = \max(\text{NAV}_1 \dots \text{NAV}_t)$$

$$\text{Drawdown}_t = (\text{NAV}_t - \text{Peak}_t) / \text{Peak}_t$$

$$\text{MaxDrawdown} = \min(\text{Drawdown}_t)$$

4. CAGR:

$$\text{Years} = \text{Days} / 365.25$$

$$\text{CAGR} = (\text{NAV_end} / \text{NAV_start})^{(1 / \text{Years})} - 1$$

5. Sortino Ratio:

$$\text{RF_daily} = (1 + \text{RF})^{(1 / 252)} - 1$$

$$D_t = \min(0, R_t - \text{RF_daily})$$

$$\text{DownsideDeviation} = \sqrt{(\sum D_t^2 / (N - 1)) \times \sqrt{252}}$$

$$\text{Sortino} = (\text{CAGR} - \text{RF}) / \text{DownsideDeviation}$$

6. Beta:

$$\beta = \text{Cov}(R_p, R_m) / \text{Var}(R_m)$$

7. Alpha:

$$\text{Benchmark CAGR} = (\text{Index_end} / \text{Index_start})^{(1 / \text{Years})} - 1$$

$$\alpha = \text{CAGR}_f - [\text{RF} + \beta (\text{BenchmarkCAGR} - \text{RF})]$$

8. Fund Risk Score(FRS):

FRS uses ABSOLUTE SEBI thresholds (same for all categories)

$$\text{ABS_MIN_V} = 0.0; \text{ABS_MAX_V} = 0.25$$

$$\text{ABS_MIN_DD} = 0.0; \text{ABS_MAX_DD} = 0.40$$

$$\text{NormVolatility} = (\text{Volatility} - 0) / (0.25 - 0)$$

$$\text{NormDrawdown} = (|\text{Drawdown}| - 0) / (0.40 - 0)$$

$$\text{FundRiskScore} = 0.4 \times \text{NormVolatility} + 0.6 \times \text{NormDrawdown}$$

9. Fund Quality Score (FQS):

$$\text{NormAlpha} = \alpha - \alpha_{\min} / \alpha_{\max} - \alpha_{\min}$$

$$\text{NormSortino} = S - S_{\min} / S_{\max} - S_{\min}$$

$$\text{NormExpense} = E - E_{\min} / E_{\max} - E_{\min}$$

$$\text{FQS} = 0.5 * \text{Norm_Alpha} + 0.4 * \text{Norm_Sortino} + 0.1 * (1 - \text{Norm_Expenseratio})$$

Layer 3 – Fund Selection and Allocation Engine

1. Risk Mismatch:

Risk mismatch measures the difference between the user's risk tolerance and the inherent risk of a mutual fund.

$$\text{RiskMismatch}_i = |\text{UserRiskScore} - \text{FundRiskScore}_i|$$

Lower values indicate better compatibility between the investor and the fund.

2. Duration Fit:

This measures how well the fund's ideal investment duration matches the user's investment horizon.

$$\text{DurationFit}_i = 1 - |\text{UserDuration} - \text{FundIdealDuration}_i| / \text{FundIdealDuration}_i$$

The value is clamped between 0 and 1. A value closer to 1 means the fund aligns well with the user's investment horizon.

3. Final Fund Score:

Each fund receives a final ranking score based on three components:

$$\text{FinalScore}_i = W_{\text{fqs}} * \text{FundQualityScore}_i - W_{\text{rm}} * \text{RiskMismatch}_i + W_{\text{df}} * \text{DurationFit}_i$$

Where:

W_{fqs} = Weight for Fund Quality Score

W_{rm} = Weight for Risk Mismatch

W_{df} = Weight for Duration Fit

Funds are sorted in descending order of FinalScore.

Portfolio Allocation Engine

4. Concentration Risk Cap

To prevent over-exposure to a single fund, the system limits the maximum capital that can be allocated to one fund.

$$\text{EffectiveCap}_i = \min(\text{MaxCapPerFund}, \text{TotalInvestment} \times \text{ConcentrationLimit})$$

Where:

$\text{MaxCapPerFund} = ₹10 \text{ Crores}$

$\text{ConcentrationLimit} = 25\%$

Example:

If $\text{TotalInvestment} = ₹1,00,000$

$\text{EffectiveCap} = \min(10\text{Cr}, 25,000) = ₹25,000$

Thus no fund receives more than ₹25,000.

5. Minimum Funds Needed:

The system determines the minimum number of funds required to safely distribute the capital.

$$\text{MinFundsNeeded} = \text{ceil}(\text{TotalInvestment} / \text{EffectiveCap})$$

This ensures that the capital can be deployed without violating concentration caps.

6. Dynamic Basket Size:

The optimizer dynamically adjusts the number of recommended funds based on the investment size.

Investment $\leq$ 20 Lakhs $\rightarrow$ 4 Funds

Investment $\leq$ 1 Crore $\rightarrow$ 5 Funds

Investment $\leq$ 5 Crores $\rightarrow$ 6 Funds

Investment $>$ 5 Crores $\rightarrow \max(8, 2 \times \text{MinFundsNeeded})$

This ensures diversification for high-net-worth investors.

7. Initial Raw Allocation:

The initial capital distribution is proportional to the final scores.

$\text{RawAllocation}_i = (\text{FinalScore}_i / \sum \text{FinalScore}) \times \text{TotalInvestment}$

8. Precision Rounding:

Allocations are rounded to the nearest ₹10 to ensure realistic investment amounts.

$\text{RoundedAllocation}_i = \text{Round}(\text{RawAllocation}_i / 10) \times 10$

9. Minimum Investment Constraint:

Each mutual fund has a minimum investment requirement.

If $0 < \text{RoundedAllocation}_i < \text{MinInvestment}_i$

$\text{Allocation}_i = \text{MinInvestment}_i$

Otherwise

$\text{Allocation}_i = \text{RoundedAllocation}_i$

10. Iterative Waterfall Allocation:

If any fund reaches its effective cap, the remaining capital is redistributed among the remaining funds.

At each iteration k:

$\text{Chunk}_i(k) = (\text{Score}_i / \sum \text{Score}) \times \text{RemainingBudget}$

If

$\text{Allocation}_i + \text{Chunk}_i > \text{EffectiveCap}_i$

Then

$\text{Allocation}_i = \text{EffectiveCap}_i$

Remaining capital is redistributed among the remaining funds.

This iterative process continues until either:

- All capital is allocated
- All funds reach their allocation caps

11. Budget Constraint:

The allocation cannot exceed the remaining available capital.

$\text{Allocation}_i = \min(\text{Allocation}_i, \text{RemainingBudget})$

12. Feasibility Check:

If the remaining amount becomes smaller than the minimum investment requirement:

$\text{Allocation}_i = 0$

Such funds are excluded from further allocation.

13. Remaining Capital Redistribution:

If there is leftover money after the waterfall process:

$\text{Remaining} = \text{TotalInvestment} - \sum \text{Allocation}_i$

The remaining amount is allocated to the highest ranked fund that has not exceeded the absolute cap.

14. Final Allocation Percentage:

$\text{AllocationPercentage}_i = (\text{Allocation}_i / \sum \text{Allocation}_i) \times 100$

Outcome:

The optimizer produces a diversified portfolio of mutual funds that:

- Matches the investor's risk tolerance
- Aligns with the investment duration
- Maximizes fund quality
- Prevents concentration risk
- Ensures full deployment of the investment capital